Lunar tides across an interrupted coastal–inland optical-clock link

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Optical clocks probe gravitational potential through their relative rates.[1,2] On a coastal-inland baseline, the predicted tidal signal combines solid-Earth deformation with a substantial ocean-loading contribution. Here we examine which lunar signatures remain constrained in an interrupted strontium–ytterbium comparison over the 1,350-km Shanghai–Wuhan fibre route.[3] Six operating periods in 2026 provide 20.1 accumulated sample days. A common M2+N2 scale is 0.973 ± 0.121, with working standard errors that account separately for residual scatter and day/period deletion. The scale changes little when constant instrumental levels are assigned by period, group or segment. Separating the constituents identifies M2 as the better-constrained response; N2 is not individually resolved. Allowing separate solar-day terms in each period shifts the common lunar scale to 0.761 ± 0.205 and changes its compatibility with unit response. A whole-template amplitude learned from other periods also reduces held-out residual variance. The comparison complements contemporaneous European network tracking[4] by testing a load-sensitive coastal–inland prediction across interrupted operating periods. It distinguishes the response supported by the observations from the additional calibration and environmental information required to identify weaker components.

The gravitational redshift links the rate of an atomic clock to its gravitational environment. Optical clocks have enabled laboratory tests of this relation, centimetre-scale geodesy and resolution of a potential gradient across a millimetre atomic sample.[5,6,7] Transportable clocks and coherent optical networks extend these measurements beyond one laboratory.[8,9,10] Time-variable potential adds a complementary observable: astronomical forcing, deformation of the Earth and redistribution of ocean water change the rates of clocks at geographically separated sites.[11,12,13]

Coastal and inland clocks sample different combinations of astronomical forcing and the Earth's response. Ocean mass redistribution adds a direct gravitational contribution and deforms the land supporting the clocks. A coastal–inland comparison therefore provides a physical test in which loading accounts for a substantial part of the prediction, rather than a small correction to an otherwise complete solid-Earth signal.[18,19,20,21] The challenge is to retain this time-dependent information when the absolute frequency level changes between operating periods.

We study this problem on a coastal–inland baseline connecting a Sr clock in Shanghai to a Yb clock in Wuhan. The related frequency-ratio experiment and the evaluations of the clock systems provide the apparatus context.[3,14,15] Long-distance coherent dissemination has been demonstrated in several net-

work configurations; published performance on other links is not assigned to this record.[16,17] According to the experimental team, the observations were acquired independently of the contemporaneous European network measurements of Stahl and colleagues.[4] This analysis was developed with knowledge of their work. We ask which part of the predicted coastal–inland tidal response remains constrained across interruptions, and how that answer changes with level resolution and solar-day freedom. The data extend overlapping summer observations with other 2026 operating periods; they are not a second independent experiment on the same link.

A phase-diverse record with measured support

Six operating periods between 7 January and 26 August 2026 contain 1,737,073 retained nominal one-second samples in 49 effective intervals. Their 2,905 non-overlapping 600-s averages span 231 calendar days and accumulate 20.1 sample days (Fig. 1a). Source membership is preserved, and each theoretical average uses the same members as its observation. These exposures describe repeated measurements of one clock comparison, rather than uninterrupted operation or six independent experiments.

The 2025 commissioning observations used different frequency conversion and processing conventions, with unresolved level corrections. They are treated separately within their September and October stages and do not enter the common 2026 response. This scope follows the experimental history rather than the size of a fitted amplitude.

The prediction combines solid-Earth and ocean-loading contributions under a documented IERS-based construction and FES2022 loading inputs.[18,19,20,21] On the retained 2026 sampling, the demeaned root-mean-square contributions are 3.245×10^{-18} and 1.507×10^{-18}, with a combined value of 4.410×10^{-18}. The loading-to-total RMS ratio is 34.2%; it is not a variance fraction because the contributions are correlated. The experiment tests the combined prediction, not separate estimates of load response or Love numbers.

The approximately 12.42-h M2 period advances through the solar day. Its changing phase relative to an exact 12-h oscillation supplies information that is absent from a single short observation window. Figure 1b shows the actual local-hour and astronomical-phase coverage. A large calendar span alone does not guarantee separation: allowing a disturbance amplitude and phase to reset frequently removes part of the information supplied by this evolution.

A static physical response across operating periods

Our reference model estimates one constant tidal scale and one constant level in each recorded operating period:

$$y_i = A_T T_i + c_{p(i)} + \epsilon_i.$$

Here T_i is the fixed, member-matched tidal prediction. The six c_p account for unknown constant levels; they are not independently measured calibration states. All parameters are fitted simultaneously to the unchanged observations. There is no preliminary alignment of segment centres to the tide, no time-dependent fitted tidal amplitude and no fit-spliced stability calculation.

The resulting scale is $A_T = 0.913 \pm 0.105$. The quoted working error is the largest of residual-scaled least-squares, delete-BJT-day and delete-period errors, reported separately in Extended Data Table 1. A conditional six-period sign test gives $P = 0.03125$ against zero and $P = 0.5625$ against unit response. The first value reaches the resolution limit of the 64-pattern, two-sided enumeration; it is not a Gaussian detection significance or an experimentally calibrated tail probability.

Replacing period levels by operating-group or individual-segment levels gives whole-template scales of 0.885 ± 0.109 and 0.832 ± 0.120. These changes test how the response depends on level freedom. They do not certify that missing corrections are constant within a period. The period reference was fixed for this revision after inspection of earlier analyses, rather than prospectively preregistered.

The response also transfers between periods. Training the scale on five periods and fitting only the nuisance level in the held-out period improves residual sums of squares in five of six periods, with the sixth changing by less than 0.02% (Fig. 3b). The pooled reduction is 3.53%, and the six training scales span 0.864–1.002. These are conditional amplitude-transfer tests: the held-out level is fitted locally and the training sets overlap.

Lunar amplitude and phase

We split the deterministic prediction into M2+N2, S2+K2 and a diurnal group, preserving the predicted constituent proportions within each group. The remaining theory is retained once as a fixed remainder. With the same six period offsets, the lunar coefficient is

$$A_L = 0.973 \pm 0.121.$$

The near-12-h and diurnal coefficients are 1.005 ± 0.271 and 0.462 ± 0.415. K2 is lunisolar, so the near-12-h coefficient is not a pure solar-gravity measurement. Changing the offset resolution gives lunar scales of 0.945 ± 0.123 for operating groups and 0.952 ± 0.137 for effective segments (Fig. 2a).

An exploratory separation of M2 and N2 clarifies what supports the common lunar response. The period-offset estimates are $A_{M2} = 0.997 \pm 0.123$ and $A_{N2} = 0.541 \pm 0.648$ (Extended Data Table 4). The stronger M2 component agrees with the expected response; N2 is not separately resolved. This motivates a physical sequence of validating the dominant prediction before examining weaker components. Correctly projected subtraction and joint estimation give the same result, retaining uncertainty in the fitted strong component.

A two-coefficient lunar fit tests phase without shifting the observation times. The in-phase and quadrature estimates are $(0.972, -0.070)$ (Fig. 2b). The unit, in-phase prediction is consistent with the working-model regions. The corresponding two-parameter six-period test gives $P = 0.03125$ against zero and $P = 0.90625$ at the unit, in-phase point. The phase plot and scalar estimate use the same observations and are not independent confirmations.

Solar-day sensitivity is assessed by adding exact 24-h and 12-h sine/cosine pairs to the joint model. A single pair set shared across the record gives $A_L = 1.023 \pm 0.123$; independent pairs in each period give 0.761 ± 0.205. The latter design retains 38.6% of the lunar information in the period-offset reference. More frequent ten-day and five-day resets give 0.891 ± 0.282 and 0.893 ± 0.430. Their scalar period-test probabilities are 0.0625 and 0.09375, whereas the period-daily model reaches 0.03125. Compatibility with unit response is also model-dependent: the period-specific daily scalar model gives $P = 0.03125$ at unity, whereas the corresponding joint phase test gives $P = 0.125$. Thus zero rejection does not by itself establish unit-response agreement. These probabilities need not vary monotonically with parameter count, because the restricted residuals and studentizing covariance also change.

The physical interpretation is specific. An unconstrained exact 12-h disturbance cannot be separated from a pure S2 response using the same frequency series alone. M2 and N2 have different frequencies and retain measurable template directions, although short-window daily freedom removes information. The free daily coefficients therefore describe a specified sensitivity class, not measured environmental states. Their inclusion is neither proof of contamination nor a mathematical violation; omitting them also leaves an assumption about unmodelled variation. Independent bounds on clocks and unmonitored transfer branches are needed to turn these tests into a complete systematic-error budget.

Physical scope and complementary measurements

The predesignated B2 round-trip monitor provides a separate check on its recorded transfer channel. On the previously validated simultaneous subset and ten-day design, adding that channel changes the whole-template scale by 0.0057 ± 0.0073. This control uses a different support and nuisance design from the new six-period reference; its numerical bound is not transferred to that reference. It cannot exclude unmonitored clock effects or laboratory fibre branches, and no same-site null comparison is available.

The reference-model response is consistent with the expected tidal scale and remains similar across the recorded group and segment level partitions. It complements network tracking through an alternative geographical baseline and an explicit accounting of interrupted support. We test a constant response to a deterministic prediction; a time-varying tidal response is outside the scope of this analysis.

The common response is not a measurement of seasonal potential or an absolute connection between operating periods. Slow signals require calibrated levels that the short-period offsets intentionally discard.

Similarly, agreement with a combined template does not independently verify every loading component, station coordinate or frequency-dependent response. The associated frequency-ratio preprint reported a different whole-template estimator on overlapping summer observations; its $0.87(15)$ and reported 5.7σ are not significance statements for the present lunar fit.[3] The result here is a reproducible, model-conditional response to a fixed tidal prediction, with the information cost of additional disturbance freedom directly exposed.

References

1. Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E. & Schmidt, P. O. Optical atomic clocks. *Reviews of Modern Physics* **87**, 637–701 (2015). Source
2. Mehlstäubler, T. E., Grosche, G., Lisdat, C., Schmidt, P. O. & Denker, H. Atomic clocks for geodesy. *Reports on Progress in Physics* **81**, 064401 (2018). Source
3. Hou, L. et al. Yb/Sr optical clock comparison at 2.2×10^{-18} uncertainty over a 1350-km fiber link. *arXiv*, 2610.34250v1 (2026). Source
4. Stahl, K. et al. Optical clock network tracks tidal variations of the gravity potential. *arXiv*, 2610.39635v1 (2026). Source
5. Chou, C. W., Hume, D. B., Rosenband, T. & Wineland, D. J. Optical Clocks and Relativity. *Science* **329**, 1630–1633 (2010). Source
6. McGrew, W. F. et al. Atomic clock performance enabling geodesy below the centimetre level. *Nature* **564**, 87–90 (2018). Source
7. Bothwell, T. et al. Resolving the gravitational redshift across a millimetre-scale atomic sample. *Nature* **602**, 420–424 (2022). Source
8. Grotti, J. et al. Geodesy and metrology with a transportable optical clock. *Nature Physics* **14**, 437–441 (2018). Source
9. Lisdat, C. et al. A clock network for geodesy and fundamental science. *Nature Communications* **7**, 12443 (2016). Source
10. Riehle, F. Optical clock networks. *Nature Photonics* **11**, 25–31 (2017). Source
11. Voigt, C., Denker, H. & Timmen, L. Time-variable gravity potential components for optical clock comparisons and the definition of international time scales. *Metrologia* **53**, 1365–1383 (2016). Source
12. Denker, H. et al. Geodetic methods to determine the relativistic redshift at the level of 10^{-18} in the context of international timescales: a review and practical results. *Journal of Geodesy* **92**, 487–516 (2018). Source
13. Schröder, S., Stellmer, S. & Kusche, J. Potential and scientific requirements of optical clock networks for validating satellite-derived time-variable gravity data. *Geophysical Journal International* **226**, 764–779 (2021). Source
14. Jia, Z.-P. et al. Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}. *Metrologia* **63**, 025002 (2026). Source
15. Zhu, Q. et al. Ytterbium lattice clock with systematic uncertainty of 1.3×10^{-18} and instability at the 10^{-19} level. *arXiv*, 2606.10514v2 (2026). Source
16. Chen, F.-X. et al. 10^{-21}-Level optical frequency dissemination over 2067 km of noise-loaded field-deployed fiber network. *Light: Science & Applications* **15**, 276 (2026). Source
17. Schioppo, M. et al. Comparing ultrastable lasers at 7×10^{-17} fractional frequency instability through a 2220 km optical fibre network. *Nature Communications* **13**, 212 (2022). Source
18. Petit, G. & Luzum, B. (eds) IERS Conventions (2010). *IERS Technical Note* **36** (2010). Source
19. LEGOS, NOVELTIS & CLS FES2022 (Finite Element Solution) Ocean Tide. *CNES data set* (2024). Source
20. Farrell, W. E. Deformation of the Earth by surface loads. *Reviews of Geophysics* **10**, 761–797 (1972). Source
21. Han, D. & Wahr, J. The viscoelastic relaxation of a realistically stratified earth, and a further analysis of postglacial rebound. *Geophysical Journal International* **120**, 287–311 (1995). Source
22. Kalman, R. E. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering* **82**, 35–45 (1960). Source

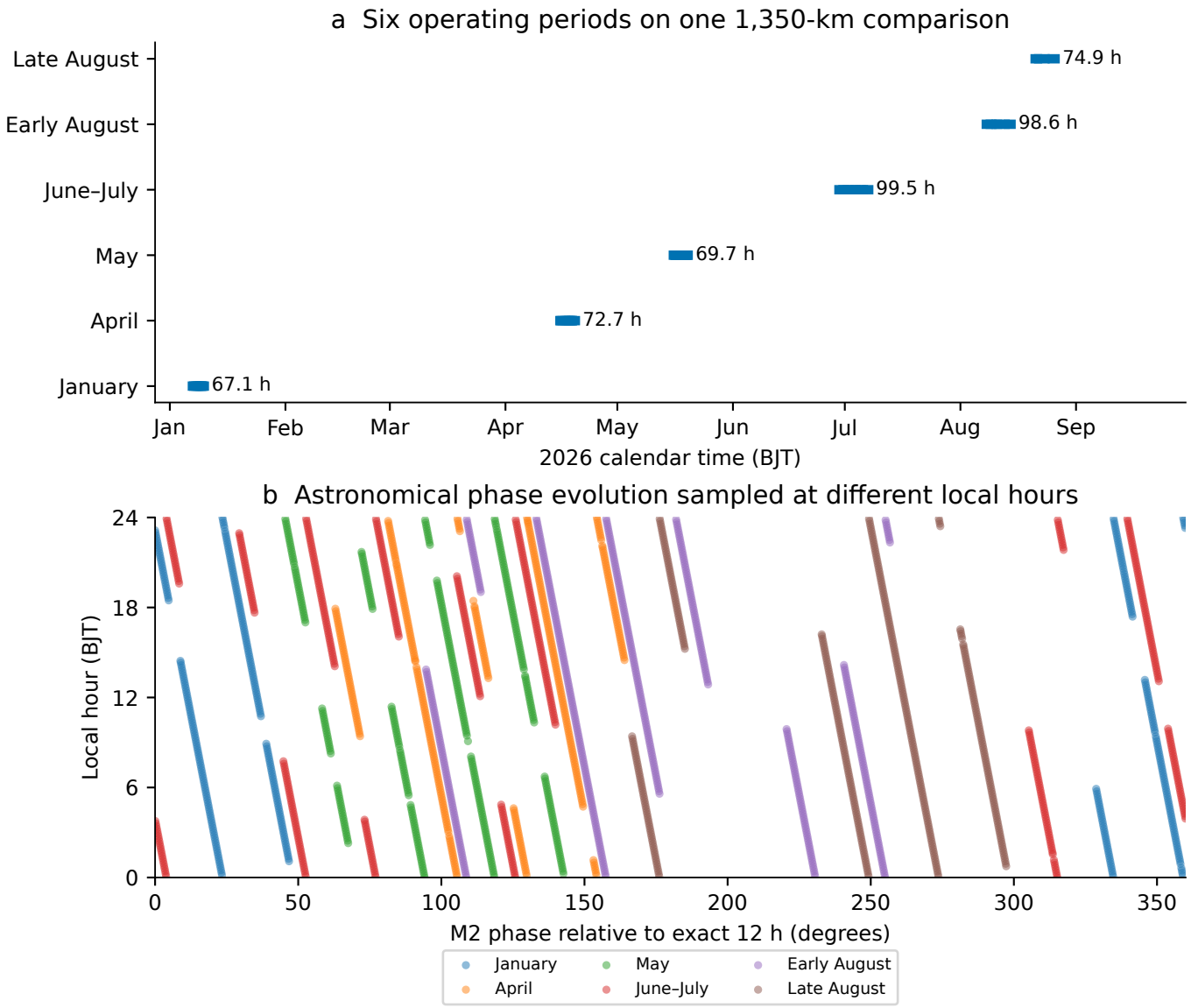


Figure 1: Observation support and astronomical phase coverage.

Fig. 1 | Observed support on the coastal-inland comparison. **a,** Recorded effective intervals within six 2026 operating periods; annotations give retained nominal sample-hours, not uninterrupted duration. **b,** Local hour versus the M2 phase relative to an exact 12-h oscillation, evaluated from the same averaged basis columns as the data analysis. Each point is one of 2,905 averages. The panels describe sampling before measurement fitting and do not independently establish separation of environmental signals.

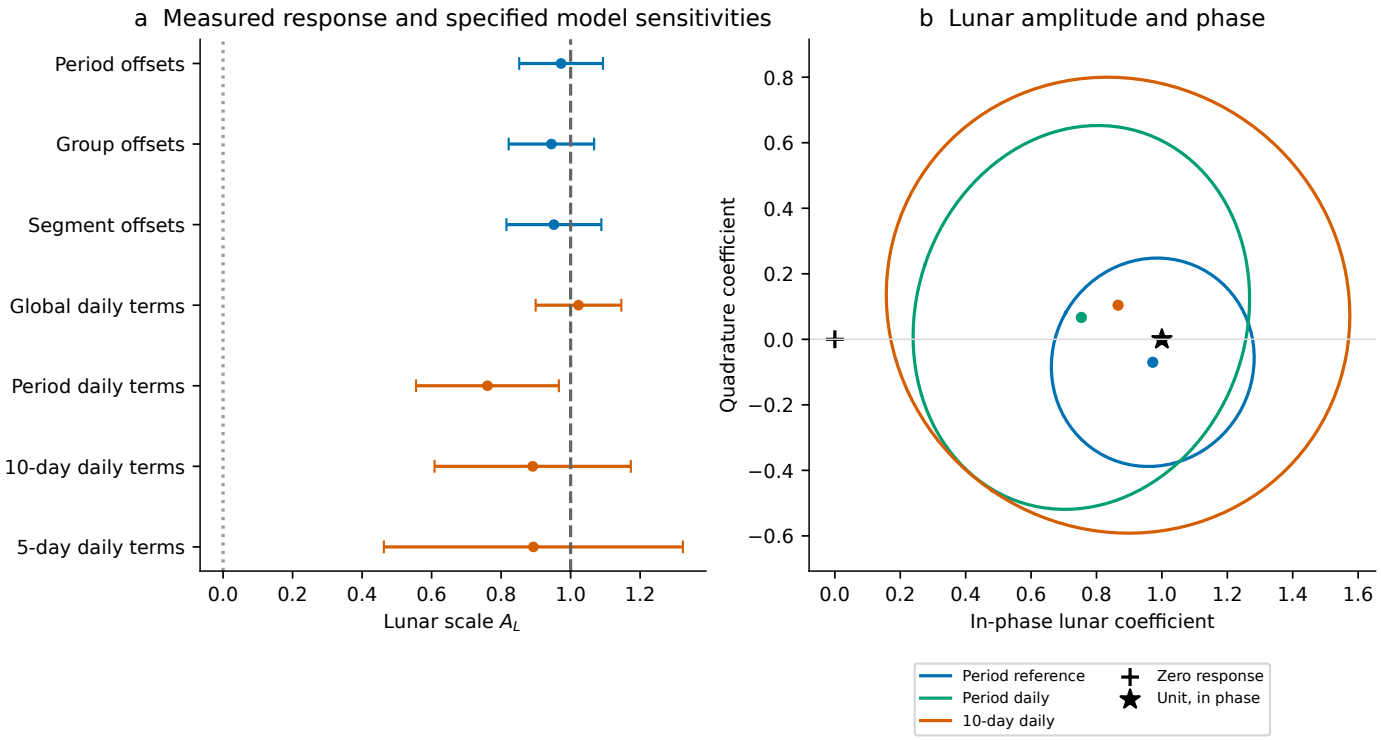


Figure 2: Lunar scale sensitivities and phase contours.

Fig. 2 | Lunar response under specified level and disturbance models. a, M2+N2 scales on identical observations. Bars are one working standard-error summary, max(LS, delete-day, delete-period), not calibrated 95% confidence intervals. The first three rows change only level resolution; later rows introduce specified daily freedom. **b,** In-phase and quadrature lunar coefficients. Curves are nominal 95% delete-day F regions with 34 denominator degrees of freedom. The unit prediction lies within all displayed regions; zero lies outside these day-based contours. This does not imply identical period-level inference: the corresponding joint sign-test probabilities at zero are 0.03125, 0.03125 and 0.25 for the period reference, period-daily and ten-day-daily models. All use the same observations, with no time shifts.

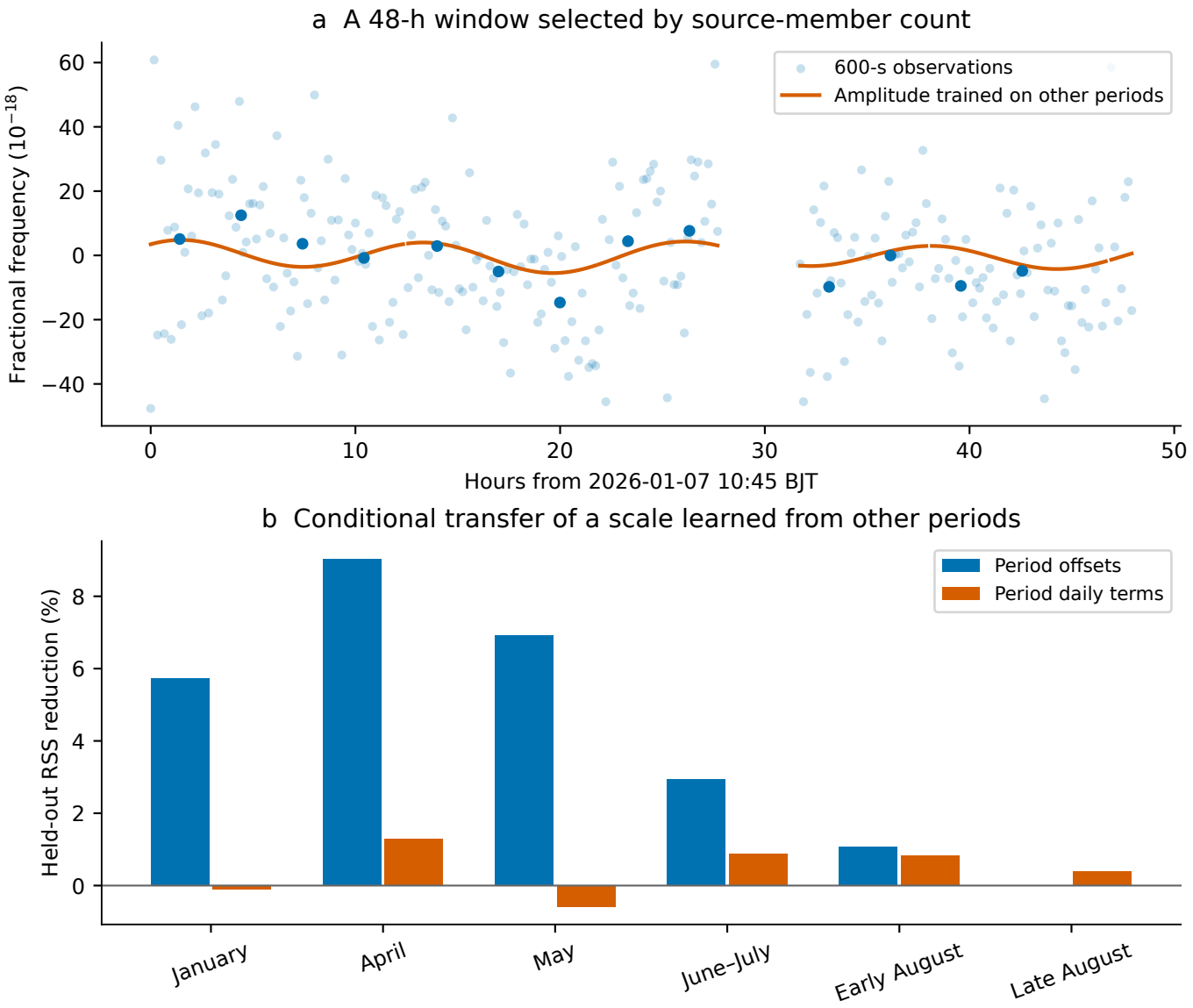


Figure 3: Representative window and held-out transfer.

Fig. 3 | A learned scale tested on held-out periods. a, The 48-h window with the largest retained source count: 158,072 members in 264 averages. Pale points show 600-s observations; dark points average 18 consecutive bins within uninterrupted effective intervals. The orange curve uses the template scale learned from the other five periods. A single level fitted across the held-out January period is removed for display; no segment centres are independently shifted. **b,** Held-out RSS reductions relative to the same locally fitted nuisance model with zero tidal scale. Negative bars are retained. The six folds share training data and supply conditional amplitude-transfer diagnostics, not independent detections.

Methods

Physical quantities, nuisance roles and model review

The observable is $y = (R/R_{\mathrm{ref}} - 1)10^{18}$, with $R = f_{\mathrm{Yb,WUH}}/f_{\mathrm{Sr,SHA}}$ and $R_{\mathrm{ref}} = 1.2075070393433377166$. Positive y means that Wuhan Yb is faster relative to Shanghai Sr. For positive Newtonian potential U, the gravitational contribution is $-[\delta U_{\mathrm{WUH}} - \delta U_{\mathrm{SHA}}]/c^2$. A template scale tests a specified prediction; it is not identified with an independently measured Love number or with a stochastic trajectory.

The revised library records every design column's definition, source and role: physical-template scale, calibration nuisance or empirical sensitivity. Calibration indicators represent constant unknown levels within recorded operating periods, groups or segments. Their interpretation is conditional on within-group constancy, not on a claim of independently calibrated levels. Empirical daily columns are confined to sensitivity models. An unspecified dynamic law is rejected by the static-model interface; adopting one would require a separately justified state, transition and noise model. A state-space estimator may also be appropriate when its transition, observation and noise laws are justified.[22] These checks enforce an explicit modelling contract, not proof that all experimental assumptions hold.

The reference design was specified during this revision after earlier model comparisons. All twenty declared reference, component and sensitivity designs are retained; no model was selected by proximity to unity or smallest probability. The reference uses the six recorded 2026 operating periods. Group, segment and fixed-calendar partitions test its dependence on level resolution. There is no underlying requirement that every statistical nuisance have a separate first-principles equation, but such a term cannot be promoted to an observed physical process.

Frequency conversion and frozen observations

The chain uses $N_{1397} = 858456$, $N_{1550} = 773598$, $N_{1156} = 1295739$ and $N_{1550,\mathrm{WH}} = 966996$. The recorded 2025 relationship gives

$$\alpha = \frac{10.93629485}{25N_{1397}}, \qquad f_{1397} = \frac{f_{\mathrm{Sr}}}{2(1-\alpha)}, \qquad f_{1550,\mathrm{WH}} = \left(N_{1550} + \frac{8}{25}\right)\frac{f_{1397}}{N_{1397}}.$$

The eight reference units include seven on transmission and one at Wuhan. For recorded counter value q in nominal hertz realized by the Wuhan reference,

$$R_{\mathrm{lock}} = \frac{N_{1156}}{N_{1397}}\frac{N_{1550} + 8/25}{N_{1550,\mathrm{WH}} + 26/20 + dq/(20\times 10^7)}\frac{1}{1-\alpha}.$$

The inherited 2026 configuration uses $d = 1$ and $\alpha = 0$, whereas the separately analysed 2025 records use $d = 2$ and the stated α. For signed Sr and Yb corrections a, b, the reconstruction uses $R =$

$R_{\mathrm{lock}}(1+a)/(1+b)(1+\Delta G)$ with inherited $\Delta G = -3.116\times10^{-15}$. Decimal anchors precede conversion to floating point; full ratios are not subtracted in float64.

The corrected experimental intervals and existing theory-independent local screen are frozen. The parent source selection contains 1,966,908 rows after gross validity screening; a six-scale running-median/MAD rule removes local excursions and neighbouring 300-s blocks, without using tidal residuals. Half-open 600-s averages remain within effective intervals and require at least 300 members. The 2026 subset contains 1,737,073 members in 2,905 averages. Observations and theory have identical source membership. A separate raw-counter readback previously checked these means, and its identities and SHA256 records accompany this revision. The new fit reads that independently reconstructed response, not any fit-adjusted series.

Unknown clock corrections remain unknown rather than measured zero. Early Sr AC entries and the May Yb correction lack complete calibration. A 37.2-mHz feedback entry at the assumed 698-nm optical node corresponds to 86.6119×10^{-18} in the affected early periods; a constant contribution is absorbed by an intercept, but time dependence is not. The historical Yb correction near -1.1×10^{-17} is not added again without resolving overlap with the updated ledger. No correction, mask or observation was changed to improve this revision's agreement.

Deterministic tidal synthesis and its limits

The frozen theoretical series combines an IERS-based solid-Earth calculation and FES2022 loading inputs.[18,19,23] Pysolid radial displacements are converted to fractional frequency through

$$s(t) = -\frac{1+k_2-h_2}{h_2}\,\frac{g[u_{\mathrm{WUH}}(t)-u_{\mathrm{SHA}}(t)]}{c^2},$$

with $h_2 = 0.6078$, $k_2 = 0.3010$ and $g = 9.794\ \mathrm{m\,s^{-2}}$. This is a displacement-to-potential approximation, not a complete degree- and frequency-dependent potential calculation. The archived loading construction includes direct load potential, induced potential and displacement through load-Love responses.[20,21] The eighteen-constituent loading input does not supply a complete long-period load model; atmospheric loading has not been independently modelled or bounded. The documented station coordinates are theory-input coordinates rather than newly certified clock-transition locations.

The 90-s theory grid is decomposed into quadratures of Q1, O1, P1, K1, N2, M2, S2, K2, Mf, Mm, Msf, Sa and Ssa plus a constant, retaining the remainder. Astronomy uses the recorded FES convention and TT−UTC=69.184 s; Beijing time is converted to UTC once.[24] Decomposition is applied to the deterministic theory, not learned from clock residuals. Interpolation precedes source-member averaging. The full model scales the total series. Component models jointly fit M2+N2, S2+K2 and K1+O1+P1+Q1, with the remaining prediction subtracted exactly once.

Earlier DE430 geometric-potential and finite-basis frequency-response comparisons supply named sensitivity checks, not certification of all template errors.[25] Their published numerical shifts belong to their original designs and are not copied as uncertainty bounds for the new period reference. The reported template scale remains conditional on the frozen physical construction. In particular, no unexplained difference between loading implementations is repaired by conjugating coefficients or tuning amplitude against the observed result.

Joint estimation and uncertainty

All fits use equal weights for the retained 600-s averages, consistent with this reduction route. They do not import the colleague P8 route's period-dependent Allan weights. Columns are normalized for rank checks and QR solution; reported coefficients use their original units. SVD supplies an independently implemented numerical check. Rank loss or numerical near-degeneracy is reported rather than resolved with an arbitrary physical amplitude: the smallest singular value of the column-normalized design must exceed 10^{-11} times the largest. Zero nuisance columns after deletion are removed; loss of a target direction is an error.

For $y = L\theta + Z\gamma + \epsilon$, partial regression is implemented with both $M_Z y$ and $M_Z L$, where $M_Z = I - ZZ^{+}$. This is algebraically equivalent to the full joint fit.[26] Its noise covariance is $M_Z \Sigma M_Z^{\mathsf{T}}$; a residual is not a new independent observation. Least-squares covariance uses the full-model residual degrees of freedom. BJT-day and operating-period jackknives refit all retained parameters. For G deletion groups,

$$C_{\mathrm{JK}} = \frac{G-1}{G} \sum_{g} (\hat{\theta}_{-g} - \bar{\theta})(\hat{\theta}_{-g} - \bar{\theta})^{\mathsf{T}}.$$

We retain LS, delete-day and delete-period covariances separately.[27] The largest marginal standard error is a working summary, not a joint covariance or a calibrated confidence interval. The phase contours use delete-day covariance and $2F_{2,34}(0.95)$; their coverage is conditional and is not certified by the covariance ellipse alone.

Restricted wild diagnostics refit each candidate null, enumerate all $2^6 = 64$ operating-period sign patterns and recompute the delete-period studentizer for each realization.[28] All patterns are retained, including globally positive and negative ties. The two-sided minimum is $2/64 = 0.03125$. The procedure assumes approximately independent, sign-symmetric period errors; nuisance estimation prevents interpreting it as an exact experimental randomization test. It does not adjust for exploratory model selection. Scalar and joint-phase probabilities are distinct. No new confidence set is inferred from tests at zero and unity alone.

The twenty new numerical designs were checked against an independent SVD implementation for coefficients, covariances and both null tests. Eight fixed-calendar designs reproduce the earlier v42 calcula-

tions. Unit tests cover exact S2/daily degeneracy, zero-support targets, joint/partial covariance agreement, constant-level invariance, input preservation and unsupported dynamic-state rejection. These are computational checks, not demonstrations of physical completeness or arbitrary-noise coverage.

Validation and subtraction of a dominant component

A physical analysis may first examine the dominant constituent and then ask whether weaker predictions are supported. Agreement of a strong component with its expected amplitude and phase supports that component's model; it neither certifies all frequencies nor supplies an independent, error-free calibration. A fit to the whole template already contains the weak constituents and is not an independent strong-only subtraction. Constituent membership must be explicit, with each fixed remainder included once.

For a same-data decomposition $y = Sa + Wb + Zc + \epsilon$, let $s = M_Z S$, $w = M_Z W$ and $v = M_Z y$. The strong-first calculation uses

$$w_\perp = w - s(s^\mathsf{T} s)^{-1} s^\mathsf{T} w, \qquad v_\perp = v - s(s^\mathsf{T} s)^{-1} s^\mathsf{T} v,$$

$$\widehat{b} = (w_\perp^\mathsf{T} w_\perp)^{-1} w_\perp^\mathsf{T} v_\perp.$$

The strong and calibration coefficients are then recovered by back-substitution. This is block elimination of the joint model, not freezing the strong-only fit after a significance screen. The complete linear map from the original observations propagates covariance; every day/period deletion repeats all stages. If the strong component instead comes from independent calibration, its calibration uncertainty must enter the residual covariance; such an independent amplitude/phase calibration is not assumed here.[26]

We checked this construction by separating the member-matched M2 and N2 columns, keeping the near-12-h and diurnal scales, fixed remainder and calibration design unchanged. Their demeaned template RMS values are 3.254 and 0.686×10^{-18}. In the period-offset design, their correlation after removing the common nuisance space is 0.06880; projecting M2 then retains 99.527% of the already nuisance-projected N2 energy. This fraction is not measured relative to the original, unprojected N2 template. Subtracting only the strong-only fit from the response and regressing on the unprojected weak column would attenuate the weak coefficient by $1 - \rho^2$ in this two-column setting.

Both elimination orders and full joint QR agree, with full-design SVD operators independently checked. Across the period, segment and period-daily designs, coefficient/deletion differences are below 2.4×10^{-12} and covariance differences below 2.3×10^{-12}. The tests are computational equivalence checks on the same data. This fixed-phase split does not independently test M2 phase or out-of-period prediction.

The M2/N2 split was added as an exploratory response to methodological review, with no multiplicity adjustment and no change to the common lunar reference estimate.

Solar-day tests, display and transfer

Daily sensitivity columns contain exact 24-h and 12-h sine/cosine pairs, globally or separately by operating period or occupied calendar block. They make no independent claim about the disturbance's mechanism or measured amplitude. S2 and a free 12-h sinusoid share a direction; binding S2 to K2 may leave a nonzero group direction without separately identifying S2. Near-degeneracy is evaluated through projected target information as well as the whole-design condition number.

Figure 3a chooses the 48-h window with the largest retained source-member count, irrespective of fit residuals. The amplitude is learned from other periods; one nuisance level is fitted using the held-out period. That level is subtracted for display only, with no separate segment recentering. Eighteen-bin display means never cross effective-interval boundaries or gaps exceeding 900 s. No plotted residual, phase contour or goodness-of-fit change is counted as an additional independent detection.

Held-out scores compare the same observations and locally fitted nuisance design with zero amplitude or an amplitude learned from other periods. The period-offset design gives a 3.525% pooled RSS reduction; period-daily freedom gives 0.539%, improving four of six periods. These folds share training data and do not constitute six independent replications. Their value is a conditional transfer check.

Conditional checks of the six-period test

The reported marginal error and the studentizer in the period-level test are different constructions. For the period-daily lunar scalar model, the working error is 0.20548, whereas its delete-period studentizing error is 0.11135. At unit response, the observed squared statistic is 4.61815; the largest of the 62 non-uniform sign-pattern statistics is 3.59197. Only the two globally uniform patterns reach the observed value, producing $P = 2/64$. This explains the enumeration without interpreting the working error as an inverted test interval. The restricted nuisance fit differs between zero and unit nulls.

We examined the estimator on its frozen sampling under three explicitly synthetic noise scenarios: independent Gaussian noise with standard deviation 20×10^{-18}; independent white noise with standard deviation 10×10^{-18} plus an Ornstein–Uhlenbeck process with standard deviation $\sqrt{300} \times 10^{-18}$ and 12-h correlation time; and variance-normalized Student-t_7 innovations with MA(1) coefficient -0.15 and marginal standard deviation 20×10^{-18}. MA memory restarts at effective-interval changes or gaps of at least 900 s. These parameters define test scenarios and are not claims about measured noise.

Each scenario uses 512 realizations, seed 2026110501, with true in-phase response one, quadrature zero and other nuisance coefficients zero. The last choice follows linear equivariance conditional on fixed designs and identifiable full and deletion fits, not an assertion that experimental offsets vanish. The

fixed theoretical remainder has already been removed in this simulation coordinate, as in the estimator. The same simulated errors are used across the three declared designs. Every realization retains all 64 restricted sign patterns and recomputes the period studentizer. Vectorized and directly enumerated P values agreed for the first realization of each design/scenario. The OU process evolves across actual elapsed time, including inter-period gaps, rather than being reset to enforce exact period independence.

True-value acceptance at the 5% threshold ranges from 95.12% to 97.46% (Extended Data Table 3). Monte Carlo standard errors are approximately 0.70–0.95 percentage points. These finite scenarios do not indicate pronounced undercoverage, but they do not certify non-stationary experimental noise, unmodelled corrections, template uncertainty or selection-adjusted inference. Marginal max-error intervals and joint-phase acceptance are distinct quantities. No observation, reference model or nuisance parameterization was changed following these tests.

Experimental controls and longer time scales

The archived B2 test uses unique member matching without interpolation, with the recorded gross-excursion screen. Its 1,519-bin reference subset and ten-day nuisance design differ from the six-period main analysis. The quoted change 0.0057 ± 0.0073 comes from paired day deletions on identical B2 support. Optical normalization uses 193399200000000 Hz, without assigning a one-way factor of two. The monitor constrains only its measured channel and overlap; matching sensitivity does not certify synchronization.

No long-term stability estimate obtained after fit-based offset adjustment is included. Short-period offset models cannot infer seasonal levels they have removed. Previously explored seasonal bounds use a separate level-preserving model and incomplete long-period loading; they are not new detections or limits from the present reference fit.

Data, code and assistance

The versioned review package includes the frozen Director observations, source identities, new static-model library, tests, all twenty results and figure-generation code. Numerical reproduction starts from the derived table; the earlier raw-readback report documents the preceding source check. This is an independently implemented reduction of the same experimental observations, not a new independent experiment. The colleague's P8 scale near 0.99 is not an input to this calculation and is not presented as an additional independent detection.

Author-approved access to raw data, derived observations and complete correction records remains to be specified before submission. The local review package is not a public archival deposit. Code availability likewise requires an author-approved archival/access statement. No raw data are included in the PDF-only private review upload.

OpenAI Codex assisted numerical implementation, code review, figure generation and drafting. Authors remain responsible for scientific verification, authorship, disclosures and access statements. Automated reviews are not author approval.

Methods references

23. Lyard, F. H., Allain, D. J., Cancet, M., Carrère, L. & Picot, N. FES2014 global ocean tide atlas: design and performance. *Ocean Science* **17**, 615–649 (2021). Source
24. Sutterley, T. C. et al. pyTMD: Python-based tidal prediction software, version 3.0.9 (version used in the primary analysis). Source
25. Folkner, W. M., Williams, J. G., Boggs, D. H., Park, R. S. & Kuchynka, P. The planetary and lunar ephemerides DE430 and DE431. *Interplanetary Network Progress Report* **42-196**, 1–81 (2014). Source
26. Ding, P. The Frisch–Waugh–Lovell theorem for standard errors. *arXiv*, 2009.06621 (2020). Source
27. Efron, B. & Stein, C. The Jackknife Estimate of Variance. *The Annals of Statistics* **9**, 586–596 (1981). Source
28. Cameron, A. C., Gelbach, J. B. & Miller, D. L. Bootstrap-Based Improvements for Inference with Clustered Errors. *Review of Economics and Statistics* **90**, 414–427 (2008). Source

Acknowledgements and funding

Author-confirmed acknowledgements, funding agencies and award identifiers are required before submission.

Author contributions

Author-specific contributions are to be supplied and approved by all authors.

Competing interests

Author declarations are required before submission.

Additional information

Correspondence details are to be supplied. This author-review PDF includes the Article, Methods and Extended Data; no separate Supplementary Information is required for the present version.

Extended Data

Extended Data Table 1 | All declared reference, component and sensitivity estimates. Marginal errors are working standard errors; P values are conditional two-sided six-period diagnostics. Phase rows report the in-phase marginal coefficient but use a joint two-parameter null test. They must not be read as scalar tests. All models use the same 2,905 observations.

Design	Target	Estimate	LS SE	Day SE	Period SE	P(0)	P(unit)
period whole	Whole	0.913	0.087	0.101	0.105	0.03125	0.56250
period lunar	Lunar	0.973	0.118	0.121	0.095	0.03125	0.81250
period phase	Joint phase	0.972	0.118	0.121	0.099	0.03125	0.90625
group whole	Whole	0.885	0.090	0.108	0.109	0.03125	0.40625
group lunar	Lunar	0.945	0.119	0.123	0.095	0.03125	0.53125
segment whole	Whole	0.832	0.097	0.120	0.102	0.03125	0.12500
segment lunar	Lunar	0.952	0.127	0.137	0.098	0.03125	0.59375
period whole daily global	Whole	1.062	0.111	0.128	0.158	0.03125	0.65625
period lunar daily global	Lunar	1.023	0.119	0.123	0.111	0.03125	0.81250
period whole daily period	Whole	0.706	0.174	0.215	0.106	0.03125	0.06250
period lunar daily period	Lunar	0.761	0.190	0.205	0.111	0.03125	0.03125
period phase daily period	Joint phase	0.754	0.191	0.201	0.100	0.03125	0.12500
10d whole	Whole	0.884	0.088	0.100	0.097	0.03125	0.31250
10d whole daily block	Whole	0.790	0.209	0.277	0.129	0.03125	0.28125
10d lunar daily block	Lunar	0.891	0.234	0.282	0.188	0.06250	0.56250
10d phase daily block	Joint phase	0.866	0.243	0.277	0.181	0.25000	0.71875
5d whole	Whole	0.886	0.088	0.101	0.097	0.03125	0.31250

Design	Target	Estimate	LS SE	Day SE	Period SE	P(0)	P(unit)
5d whole daily block	Whole	0.924	0.291	0.383	0.232	0.06250	0.78125
5d lunar daily block	Lunar	0.893	0.308	0.430	0.307	0.09375	0.71875
5d phase daily block	Joint phase	0.892	0.308	0.437	0.305	0.31250	0.90625

Extended Data Table 2 | Interpretation of model terms.

Term	Definition and role	Information not supplied
Deterministic tide	Prediction from adopted astronomy and Earth/load response	Independent certification of every model component
Constant template scale	Common response to a specified time series	A separately measured Love number or evolving physical state
Period/group/segment offset	Jointly estimated constant calibration nuisance	Measured absolute connection or within-period constancy proof
Exact daily sinusoids	Empirical disturbance sensitivity class	A measured mechanism, amplitude prior or unique S2 attribution
Residual covariance	Statistical description of unexplained variation	Identification of its physical source
Phase quadrature	Constant component-response diagnostic	A fitted shift of observed times or an independent experiment

Extended Data Table 3 | Synthetic conditional validation on the frozen sampling. Each scenario has 512 realizations; error units are dimensionless template scale. D1 denotes the period-reference scalar lunar fit, D2 its period-daily alternative, and D3 the two-coefficient period-daily fit. Noise codes are G, independent Gaussian; O, white plus OU; and T, Student-t plus MA. Max-SE means max(LS, delete-day, delete-period). All bias, SD and marginal-coverage columns concern the in-phase coefficient. Wild acceptance is scalar for the first two designs and joint for the third. MC SE is a binomial Monte Carlo standard error in percentage points; it is not experimental uncertainty. Noise scenarios were locally specified before these simulations, after the experimental analysis; they were not prospectively preregistered and do not certify the real noise process.

Design	Noise	Bias	SD	Mean max-SE	1.96 max-SE coverage	Wild acceptance	MC SE
D1	G	0.0031	0.1141	0.1331	96.68%	97.46%	0.70
D2	G	0.0112	0.1795	0.2328	97.07%	97.27%	0.72
D3	G	0.0111	0.1784	0.2354	97.27%	96.48%	0.81
D1	O	-0.0089	0.2400	0.3004	98.44%	96.29%	0.84
D2	O	0.0126	0.4051	0.6001	99.22%	95.70%	0.90
D3	O	0.0155	0.4068	0.6090	99.41%	95.70%	0.90
D1	T	-0.0043	0.1017	0.1212	97.66%	95.70%	0.90
D2	T	-0.0024	0.1595	0.2076	98.05%	96.48%	0.81
D3	T	-0.0025	0.1596	0.2093	98.24%	95.12%	0.95

Extended Data Table 4 | Strong/weak constituent cross-check on unchanged observations. M2 and N2 have separate constant scales; near-12-h and diurnal scales are estimated jointly, with the same fixed remainder as the grouped lunar model. Uncertainties are max(LS, delete-day, delete-period) working errors. P values are conditional, two-sided six-period sign diagnostics with minimum 0.03125, without multiplicity or model-selection adjustment. No strong coefficient is fixed after a P-value screen. N2 is not individually resolved in these designs.

Design	M2	N2	M2 P(0)	M2 P(1)	N2 P(0)	N2 P(1)
Period	0.997 ± 0.123	0.541 ± 0.648	0.03125	0.93750	0.34375	0.34375
Segment	0.942 ± 0.135	1.137 ± 0.830	0.03125	0.46875	0.15625	0.87500
Period+daily	0.820 ± 0.200	-0.043 ± 0.852	0.03125	0.28125	0.96875	0.12500